Q.1 A full adder circuit takes _____ bit(s) as input and returns _____ bit(s) as output.

A. 2, 1

B. 2, 2

C. 3, 1

D. 3, 2

Q.2 In order to build an XOR gate using NAND gates only, _____ gate(s) are required.

A. 1

B. 4

C. 2

D. 3

Q.3 Which of the following expressions represents S output in a full adder?

A. ({A B}) {C_{in}}

B. ({A {C_{in}}}) B

C. ({A + B}) {C_{in}}

D. A B {C_{in}}

Q.4 Which of the following is an invalid BCD code?

A. 1101

B. 1001

C. 0101

D. 0011

Q.5 When implementing a half adder by using gates, which of the following gates is used to give Sum (S) as an output?

A. XOR

B. NOR

C. OR

D. AND

Q.6 {\text{In }}{\mathbb{R}^n},{\text{which of the following is true about }}\phi = \left\{ {} \right\}{\text{ and }}A = \left\{ {\left({{a_1},{a_2}, \ldots ,{a_n}} \right),{a_i} \in \mathbb{R},1 \leqslant i \leqslant n,i \in \mathbb{N}} \right\}?

A. {{ is Connected and }}A{{ is Disconnected}}

B. {{ is Disconnected and }}A{{ is Connected}}

C. {{Both are Connected}}

D. {{Both are Disconnected}}

Q.7 {\text{Set of isolated point(s) of the complement of set }}\left\{ {\left({x,y} \right): - n < x,y < n,\left({x,y} \right) \ne \left({0,0} \right),n \in \mathbb{N}} \right\}{\text{ in }}{\mathbb{R}^{\text{2}}},{\text{is}} - - - - .

A. \{ {({x,y}) {{R}^2}:x = y = | n |,n {N}} \} \{ {({0,0})} \}

B. \{ {({x,y}) {{R}^2}:x = y = | n |,n {N}} \}

C. \{ {({0,0})} \}

D. \{ {({x,y}) {{R}^2}:x = y = | n |,n {N}} \} \{ {({0,0})} \}

Q.8 \begin{gathered} {\text{For a non empty closed and bounded subset }}S{\text{ in }}{\mathbb{R}^n},{\text{ if }}{\rm H}{\text{ is the collection open sets such that }} \hfill \\ S \subset \cup \left\{ {H:H \in {\rm H}} \right\},{\text{then by Heine - Borel theorem,}} \hfill \\ \end{gathered}

A. S _{ } \{ {{H_ }:{H_ } { H}} \}

B. S _{ = 1}^ \{ {{H_ }:{H_ } { H}} \}

C. S _{ = 1}^n \{ {{H_ }:{H_ } { H}} \}

D. {{All above are equally valid}}

Q.9 The limit of the function f(x) = \frac{{xy}}{{{x^2} + {y^2}}} by letting \left({x,{\rm{ }}y} \right)approach \left({0,{\rm{ }}0} \right)along the line y{\rm{ }} = {\rm{ }}x is _____.

A. - {1}{2}

B. Undefined

C. Finite

D. {1}{2}

Q.10 {\text{A compact set in }}{\mathbb{R}^n}{\text{ is - - - - - - - - - }}{\text{.}}

A. {{open and unbounded}}

B. {{closed and unbounded}}

C. {{open and bounded}}

D. {{closed and bounded}}

Q.11 {\text{In }}{\mathbb{R}^n},\,{\text{the interior }}{S^0}{\text{ of a non - empty set S is - - - - - - }}{\text{.}}

A. {{open - disconnected}}

B. {{open - connected (region)}}

Q.12 {\text{The set }}\left\{ {\left({x,y} \right): - n < x,y < n,\left({x,y} \right) \ne \left({0,0} \right),n \in \mathbb{N}} \right\}{\text{ is - - - - - - in }}{\mathbb{R}^{\text{2}}}.

A. neither open nor closed

B. both open or closed

C. closed

D. open

Q.13 \begin{gathered} {\text{In }}{\mathbb{R}^n}{\text{, if }}{S_1},{S_2},{S_3} \ldots \,\,{\text{are non - empty closed subsets such that }}{S_1} \supset {S_2} \supset \cdots \; \supset {S_r} \supset \cdots {\text{ and }} \hfill \\ {\text{sup}}\left\{ {\left| {{X_r} - {Y_r}} \right|:{X_r},{Y_r} \in {S_r},r \geqslant 1} \right\} \to 0{\text{ as }}r \to \infty {\text{,then the order of set }}\mathop \cap \limits_{r = 1}^\infty {S_r} = - - - . \hfill \\\ \end{gathered}

A. {{zero}}

B. {{one}}

C. {{finite but more than one}}

D. {infinite}

Q.14 A set " S " is polygonally connected if, ------- pair of points in S can be connected by a polygonal path lying -------- in " S ".

A. some, entirely

B. every, partially

C. some, partially

D. every , entirely

Q.15 If \mathop {\lim }\limits_{X \to {X_0}} f\left(X \right) exists, then it is ____.

A. 0

B. none of these

C. 1

D. unique
